



Department of Computer Engineering

Semester	T.E. Semester VII – CMPN
Division & Batch	CMPN C
Subject	Devops
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Experiment Number	01
Experiment Title	Understanding DevOps: Principles, Practices, and DevOps Engineer Roles & Responsibilities
Aim	To understand the concept of DevOps, its principles, practices, and the roles and responsibilities of a DevOps Engineer.
Description	Introduction DevOps is a software development methodology that combines Development (Dev) and Operations (Ops) to improve collaboration between software developers and IT operations teams. It focuses on automating the software development lifecycle, enabling faster software delivery, improved quality, and continuous monitoring. The primary goal of DevOps is to shorten the development cycle while delivering reliable and high-quality software. Principles of DevOps 1. Collaboration Development and Operations teams work together throughout the software lifecycle to achieve common goals. 2. Automation

Automating tasks such as code integration, testing, deployment, and infrastructure management reduces manual effort and human errors.

3. Continuous Integration (CI)

Developers frequently merge their code into a shared repository where automated builds and tests are performed.

4. Continuous Delivery/Deployment (CD)

Software changes are automatically prepared and deployed to production quickly and safely.

5. Monitoring and Feedback

Applications and infrastructure are continuously monitored to identify issues and improve system performance.

6. Infrastructure as Code (IaC)

Infrastructure is managed using configuration files and code rather than manual processes.

DevOps Practices

- Continuous Integration (CI)
- Continuous Delivery (CD)
- Continuous Deployment
- Infrastructure as Code (IaC)
- Version Control (Git)
- Automated Testing
- Configuration Management
- Continuous Monitoring
- Containerization (Docker)
- Container Orchestration (Kubernetes)

Common DevOps Tools

Category	Tools
Version Control	Git, GitHub
CI/CD	Jenkins, GitHub Actions, GitLab CI
Build Tools	Maven, Gradle
Containers	Docker
Orchestration	Kubernetes
Configuration Management	Ansible, Puppet, Chef
Monitoring	Prometheus, Grafana

Roles and Responsibilities of a DevOps Engineer

A DevOps Engineer is responsible for bridging the gap between development and operations teams.

	Responsibilities ● Develop and maintain CI/CD pipelines. ● Automate software build, testing, and deployment processes. ● Manage cloud infrastructure and servers. ● Monitor application performance and system health. ● Ensure system security and compliance. ● Manage containerized applications using Docker and Kubernetes. ● Troubleshoot deployment and infrastructure issues. ● Implement Infrastructure as Code (IaC). ● Collaborate with developers, testers, and system administrators. ● Optimize application performance and scalability. Benefits of DevOps ● Faster software delivery ● Improved collaboration ● Higher software quality ● Reduced deployment failures ● Faster issue resolution ● Better customer satisfaction ● Increased automation ● Improved scalability and reliability DevOps Lifecycle Plan ↓ Develop ↓ Build ↓ Test ↓ Release ↓ Deploy ↓ Operate ↓ Monitor * (Continuous Feedback)
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Conclusion	Successfully understood the fundamentals of DevOps, its core principles, common practices, widely used tools, and the roles and responsibilities of a DevOps Engineer in modern software development.
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